

Masterarbeit im Studiengang Linguistik Master of Arts
an der Heinrich-Heine-Universität Düsseldorf



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Todo list

make sure this formula works	3
More detailed instructions and notes. Possibly spanning multiple lines, so you can ölkjknu kducbejn	3
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Acknowledgements

I am grateful to my advisors for their invaluable feedback.

This work would not have been possible without my Bialetti moka pot.

Abstract

This template has been developed by David Arps (davidarps.github.io). It can serve as a guideline for Bachelor and Master theses, in particular in domains such as Computational Linguistics, Natural Language Processing, Linguistics, Computer Science. Depending on your subject, there might be different best practices, so check with your thesis advisor.

This document is compiled using **xelatex**.

1. Introduction

Motivation This template was created to make sure that students can focus on the interesting stuff (writing an interesting thesis), rather than having to set up a latex template. The template is built from my own Master thesis, which I can make available upon request. Other master theses are available via the Institute of Linguistics (<https://www.ling.hhu.de/>)

Structure of this template The introduction contains some notes about the template, and below that you will see some examples of how to add Tables, Figures, examples, etc. I chose to write the text in different tex files, because it makes the main file more readable. As a clear advantage, this accelerates writing a single chapter. E.g. the main file has blocks such as this:

```
\chapter{Linguistic Background}\label{ch:linguistic-background}  
\input{text/linguistic-background}
```

You can just comment out such a block in order to not compile this chapter.

Workflow The described techniques work well when writing locally, using VSCode and the Latex Workshop plugin as an IDE. This setup allows to version your thesis using git, e.g. with a private github repository. Overleaf projects can also be synced with git, so a hybrid local/overleaf setup is also possible.

2. Todo lists

Motivation One of the most difficult parts of longer-running projects, such as a thesis, is keeping track of what you have to do. As a rule, find a system that is simple, and that allows you to keep momentum in your most important goal, namely writing. Here is what worked for me.

TODO notes The `todonotes` latex package allows to add todos in the pdf document. This means you have the todo elements directly in your tex code, rather than overleaf comments, a third-party software, handwritten notes, etc. In the beginning of this document, there appears a Todo List which you can use as a starting point if you do not know what to do next.

Types of TODOs The most basic ones are small TODO notes on the side. For longer TODO notes, such as thoughts on how to structure a section, you can use inline todos with `\todo[inline]bla`:

make sure this formula works

More detailed instructions and notes. Possibly spanning multiple lines, so you can ölkjknu kducbejn

If you have to draw a figure, table, etc., you can use `missingfigure`.

By the way, a great way to sketch figures is quickly drawing them on paper, and then putting a picture of that drawing in your draft with `\includegraphics`. This way you defer the time-consuming digital drawing to a point in time when you know that the figure will definitely stay.

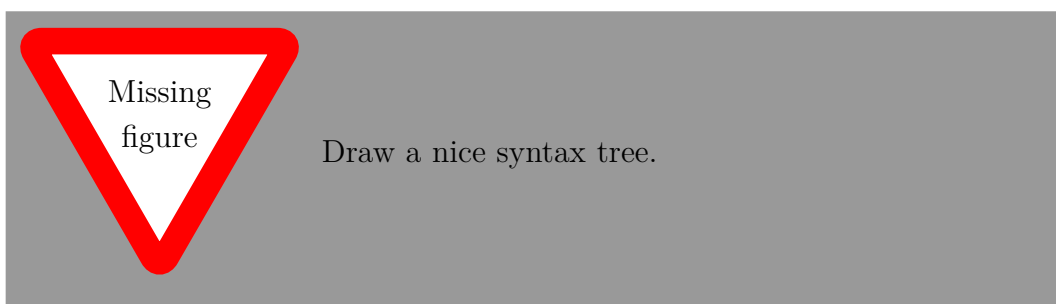


Figure 2.1.: Syntax tree

2. *Todo lists*

It is also possible to define custom todo commands. For example, you might write in your discussion a sentence such as this:

Related work has shown that trees mostly grow downwards in Linguistics (Section), and upwards in nature.

In the preamble (in `commands.tex`), the command `\findref` is defined to make a todo note in a particular color, and add the `findref` prefix so that you know you have to find a tex reference label.

Similarly, the `\findbib` command makes a todo note for searching for a literature reference, or a bibtex entry. For example like this This allows you to stay in the flow of writing, instead of interrupting this flow to look for the bibtex entry. Later, you can resolve all of these Todo items in one go.

Of course, you can define more specific todo commands if you have the need.

find ref: refer to related work section

find bib: bibtex entry for some related work

3. Example tex Code

3.1. Linguistic examples

Linguistic Examples:

The first sentence is taken from a review and part of the treebank released by Silveira et al. (2014). The second sentence is made-up.

- (1) a. The service was friendly and fast, but this just doesn't make up for the lackluster product.
- b. The popcorn was fresh and salty, but this just doesn't make up for the bad movie.
- c. The interior was outdated and fresh, but this rather doesn't make up for the newborn period.

Glossing is also possible and should follow the Leipzig Glossing rules as closely as possible.

- <https://www.eva.mpg.de/lingua/pdf/Glossing-Rules.pdf>
- <https://ctan.org/pkg/gb4e>

An example with some light glossing:

- (2) a. Der Mann sieht die Schauspielerin im
DEF.NOM.SG man see.PRES.3SG DEF actress in
Fernsehen.
TV
'The man sees the actress on TV.'
- b. Die Schauspielerin sieht der Mann im
DEF actress see.PRES.3SG DEF.NOM.SG man in
Fernsehen.
TV
'The man sees *the actress* on TV.'
- c. Die Schauspielerin sieht den Mann im Fernsehen.
'The actress sees the man on TV.'

3.2. Dependency trees

Using `tikz-dependency`, example in Figure 3.1. Code for bigger figures and tables should be put in separate files.

1. Define the set of UPOS tags that are considered content words.
2. For each content word token t in the input treebank, collect the syntactic context $s(t)$. Fill the map $\mathcal{C}$ that maps syntactic contexts to the set of word forms that appear in that context.
3. Create the output treebank. Replace each content word token t in the input treebank with a random token from the set $\mathcal{C}(s(t))$.

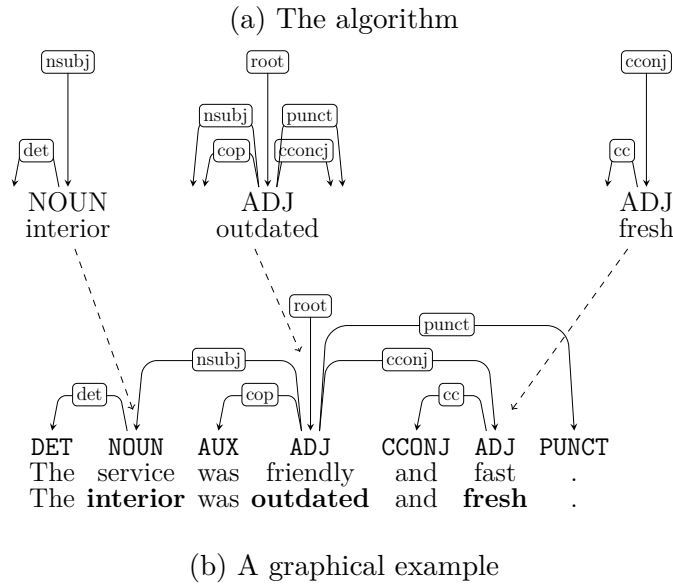


Figure 3.1.: Language-independent nonce data generation algorithm. The syntactic context $s(t)$ of the replacements is illustrated by the tree fragments. Furthermore, $\mathcal{C}$ contains the information that $s(\text{service}) = s(\text{interior})$, $s(\text{friendly}) = s(\text{outdated})$, $s(\text{fast}) = s(\text{fresh})$.

3.3. Tables

Example table is displayed in Table 3.1. For complex tables, multicolumn and multirow are your friends!

3. Example tex Code

	train	test	NOUN		PROP		ADJ		ADV		VERB		total
	tokens	tokens	freq.	repl.	freq.	repl.	freq.	repl.	freq.	repl.	freq.	repl.	repl.
ar	224K	28K	0.34	0.93	0.01	0.77	0.10	0.93	0.01	0.74	0.08	0.66	0.47
de	2754K	326K	0.21	0.96	0.06	0.93	0.08	0.68	0.06	0.92	0.08	0.61	0.41
en	204K	25K	0.16	0.85	0.08	0.84	0.07	0.86	0.04	0.83	0.11	0.59	0.38
fr	355K	10K	0.19	0.83	0.05	0.69	0.06	0.88	0.05	0.92	0.08	0.50	0.36
ru	1206K	158K	0.23	0.84	0.04	0.70	0.09	0.80	0.05	0.89	0.11	0.46	0.39
zh	99K	12K	0.28	0.88	0.08	0.92	0.02	0.80	0.05	0.94	0.15	0.65	0.48

Table 3.1.: Data statistics for Experiment 1. For each UPOS tag that is affected by the replacement, freq. denotes the relative frequency of the tag in the testing data, and repl. denotes the relative frequency of forms with that tag that are replaced with other forms. The total column shows the relative frequency of forms that are replaced across all tags. zh = Chinese.

4. Bibliography

Custom bib entries are stored in `biblio.bib` When your thesis is in CL or NLP, I strongly recommend using the anthology bib files: <https://github.com/acl-org/acl-style-files/blob/master/anthology.bib.txt>

Bibliography

Natalia Silveira, Timothy Dozat, Marie-Catherine de Marneffe, Samuel Bowman, Miriam Connor, John Bauer, and Chris Manning. 2014. A Gold Standard Dependency Corpus for English. In *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14)*, pages 2897–2904, Reykjavik, Iceland. European Language Resources Association (ELRA).

A. Appendix with more results

All the stuff that didn't make it to the main paper.

B. Declarations

B.1. Declaration of AI usage

I, _____ (*first name and last name*), Matrikelnummer _____,
hereby confirm that I have read and acknowledged the AI policy of the Institute for Linguistics, HHU (June 2025).

I hereby declare that for my term paper/thesis/project with the title _____ (*title*),
in the seminar _____ (*course name*) with _____ (*lecturer/supervisor name*)

☐ I have not used any AI tools.

☐ I have used AI. The following list contains all tools that I have used as well as a short description of how and for what purpose these tools were used.

Name of the tools	version/date	Purpose
<i>e.g., Grammarly</i>	grammarly free (2025), 2/2025	<i>spell-correction</i>
<i>e.g., ChatGPT</i>	ChatGPT-4o, 3/2025	<i>to generate code to create the plot on figure XX</i>
...		

B.2. Erklärung

Hiermit erkläre ich, dass ich die vorliegende Arbeit selbständig verfasst und keine anderen als die angegebenen Quellen benutzt habe. Die Stellen der Arbeit sowie beigefügte Zeichnungen, Skizzen, graphische Darstellungen, die anderen Werken dem Wortlaut oder dem Sinn nach entnommen sind, habe ich unter Angabe der Quellen kenntlich gemacht.

Signature at the end

Your name

Düsseldorf, den

Submission
date